计算机网络

18.

File and WWW



厦门大学软件学院

黄炜 助理教授

Ch 31 File Transfer And Remote File Access

文件传输与远程文件访问

FTP (File Transfer Protocol)

- 文件传送协议 FTP (File Transfer Protocol)
  - FTP 提供交互式的访问，允许客户指明文件的类型与格式，并允许文件具有存取权限。
  - FTP 屏蔽了各计算机系统的细节，因而适合于在异构网络中任意计算机之间传送文件。
  - RFC 959 很早就成为了因特网的正式标准。
- 文件传送并非很简单的问题
  - 众多的计算机厂商研制出的文件系统数百种，且差别很大。

FTP 特点

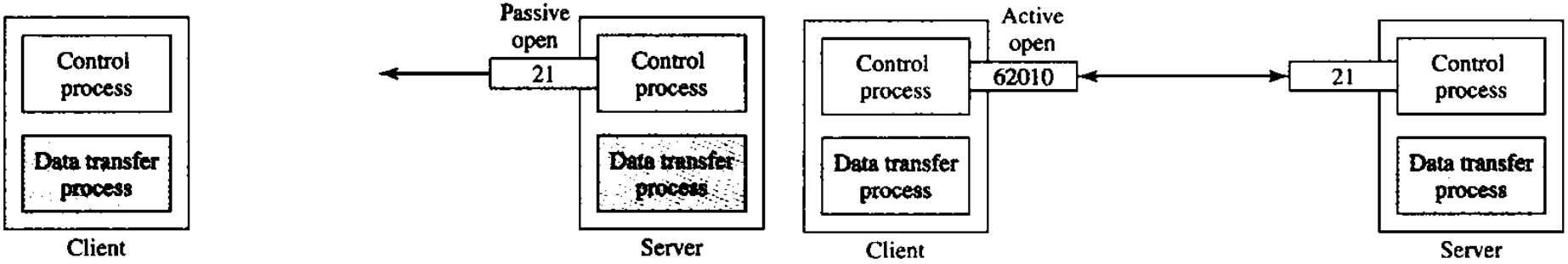
- 文件传送协议 FTP 只提供文件传送的一些基本的服务，它使用 TCP 可靠的运输服务。
- FTP 的主要功能是减少或消除在不同操作系统下处理文件的不兼容性。
- FTP 使用客户服务器方式。
  - 一个 FTP 服务器进程可同时为多个客户进程提供服务。
  - FTP 的服务器进程由两大部分组成：一个主进程，负责接受新的请求；另外有若干个从属进程，负责处理单个请求。

FTP 两个不同的端口号

- 当客户进程向服务器进程发出建立连接请求时，要寻找连接服务器进程的熟知端口(21)，同时还告诉服务器进程自己的另一个端口号，用于建立数据传送连接。
- 接着，服务器进程用自己传送数据的熟知端口(20)与客户进程所提供的端口号码建立数据传送连接。
- FTP 使用2个端口号，数据连接与控制连接不会混乱。
  - 使协议更加简单和更容易实现
  - 传输文件还可利用控制连接（如：客户发送请求终止传输）

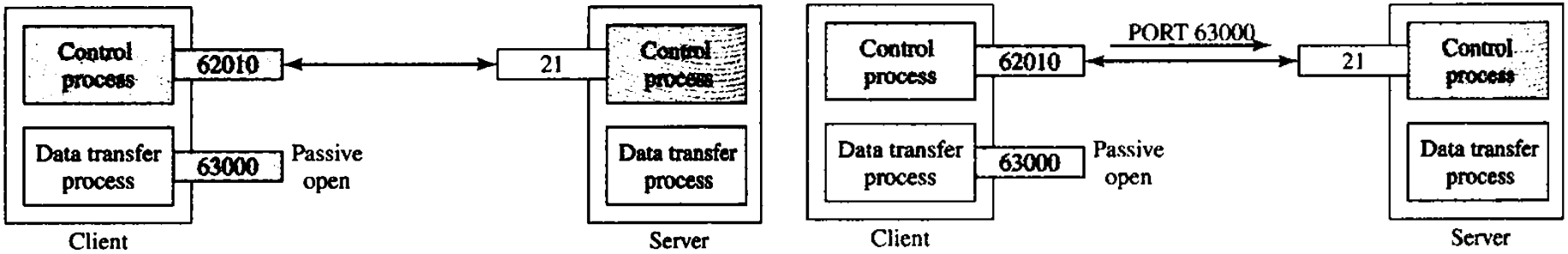
Control

- **The server issues a passive open on the well-known port and waits for a client.**
- **The client uses an ephemeral port and issues an active open.**



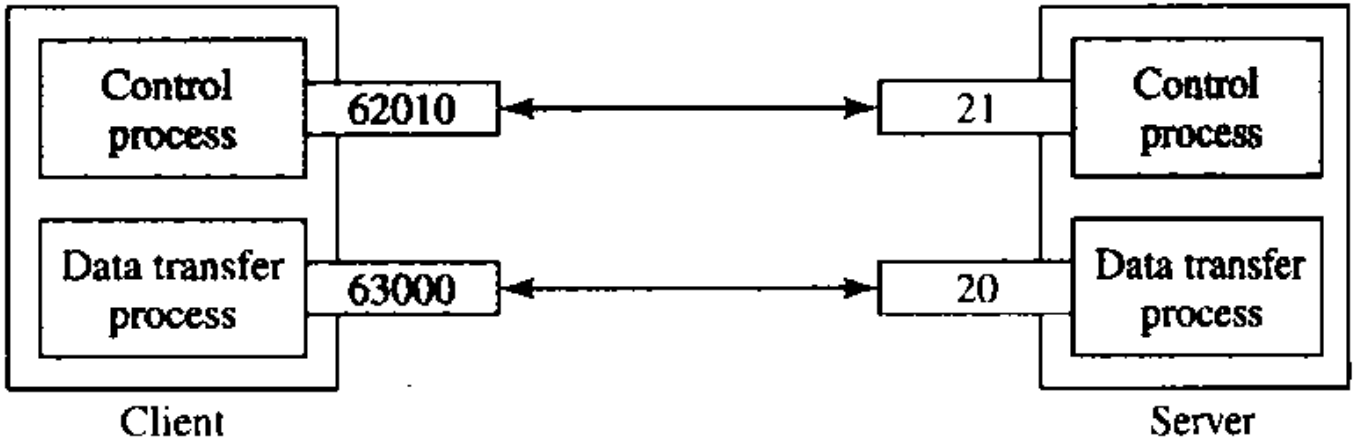
Data Connection

- **The client issues a passive open using an ephemeral port.**
- **The client sends this port number to the server using the PORT command.**



Data Connection

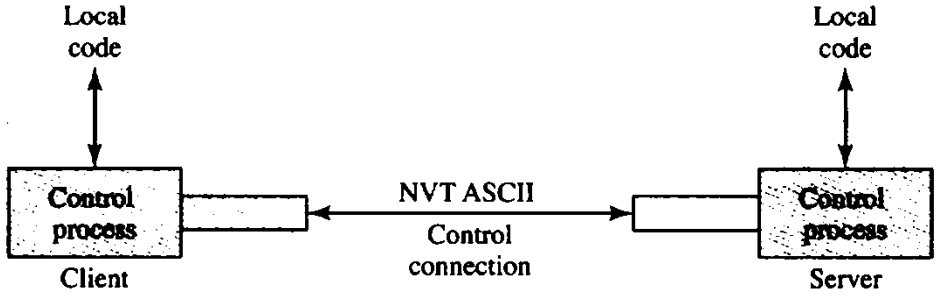
- **The server receives the port number and issues an active open using the well-known port 20 and the receive ephemeral port number.**



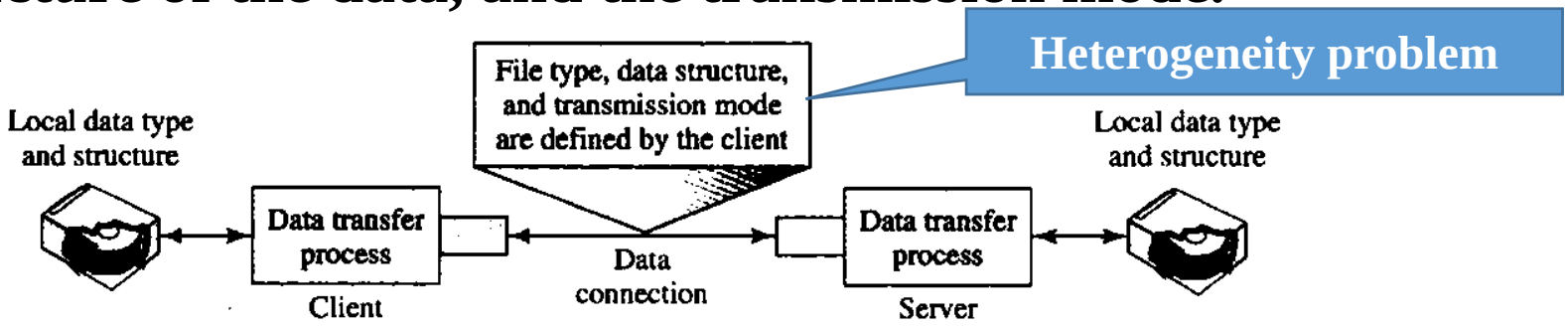
c. Active open by server

Communication over Connections

- **Communication over Control Connection**
  - **Communication is achieved through commands and responses.**



- **Communication over Data Connection**
  - **The client must define the type of file to be transferred, the structure of the data, and the transmission mode.**



File Type (1)

- **ASCII file**
  - 这是用于传输文本文件的默认格式。NVT ASCII编码。
- **EBCDIC file（扩充的二—十进制交换码）**
  - Extended Binary-Coded Decimal Interchange Code.
- **Image file:**
  - 这是用于传输二进制文件的默认格式。文件被发送为连续的比特流，没有任何解释或编码。
  - 这主要是用来传输二进制文件，如编译过的程序。

Data Structure

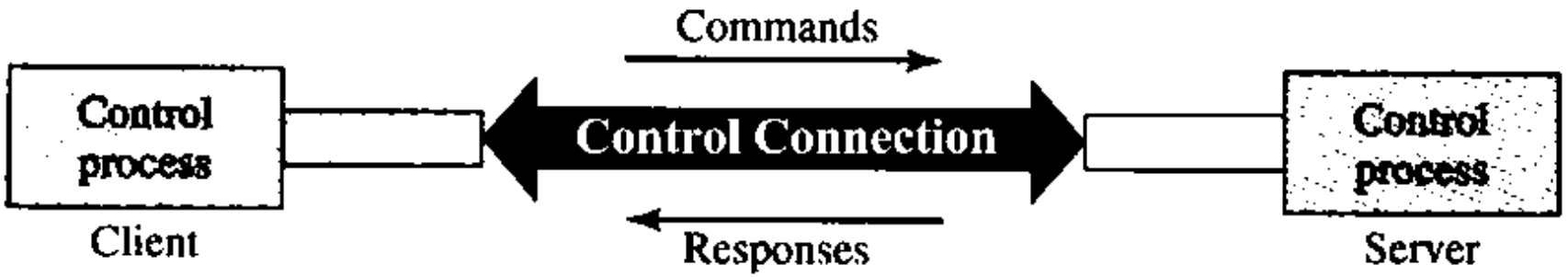
- **File structure(default):**
  - 它是一个连续的字节流。
- **Record structure:**
  - 文件划分为记录。只用于文本文件。
- **Page structure:**
  - 文件分为两页，每一页有一个页和一个页头。
  - 该网页可以随机或顺序存储或访问。

Transmission mode

- **Stream mode: (default)**
  - 数据传送从FTP的TCP作为连续的字节流。
- **Block mode:**
  - 数据可以按分块从FTP到TCP分发。每块前有3字节的头。
- **Compressed mode:**
  - 如果文件大，可以压缩数据。所使用的压缩方法通常是游程长度编码。在二进制文件中，空字符通常被压缩。

Command processing

- **FTP uses the control connection to establish a communication between the client control process and the server control process.**



FTP Commands

- **Access commands**
  - **These commands let the user access the remote system.**

Table 20.1 *Access commands*

Command	*Argument(s)*	*Description*
USER	User id	User information
PASS	User password	Password
ACCT	Account to be charged	Account information
REIN		Reinitialize
QUIT		Log out of the system
ABOR		Abort the previous command

FTP Commands

- **File management commands**

Table 20.2 *File management commands*

Command	*Argument(s)*	*Description*
CWD	Directory name	Change to another directory
CDUP		Change to the parent directory
DELE	File name	Delete a file
LIST	Directory name	List subdirectories or files
NLIST	Directory name	List the names of subdirectories or files without other attributes
MKD	Directory name	Create a new directory

Table 20.2 *File management commands (continued)*

Command	*Argument(s)*	*Description*
PWD		Display name of current directory
RMD	Directory name	Delete a directory
RNFR	File name (old file name)	Identify a file to be renamed
RNTO	File name (new file name)	Rename the file
SMNT	File system name	Mount a file system

FTP Commands

- **Data formatting commands**

Table 20.3 *Data formatting commands*

Command	*Argument(s)*	*Description*
TYPE	A (ASCII), E (EBCDIC), I (Image), N (Nonprint), or T (TELNET)	Define the file type and if necessary the print format
STRU	F (File), R (Record), or P (Page)	Define the organization of the data
MODE	S (Stream), B (Block), or C (Compressed)	Define the transmission mode

- **Port defining commands**
  - 简单的用户定义的端口号的数据连接的客户端的站点。

Table 20.4 *Port defining commands*

Command	*Argument(s)*	*Description*
PORT	6-digit identifier	Client chooses a port
PASV		Server chooses a port

FTP Commands

- **File transfer commands**

Table 20.5 *File transfer commands*

Command	*Argument(s)*	*Description*
RETR	File name(s)	Retrieve files; file(s) are transferred from server to the client
STOR	File name(s)	Store files; file(s) are transferred from the client to the server
APPE	File name(s)	Similar to STOR except if the file exists, data must be appended to it
STOU	File name(s)	Same as STOR except that the file name will be unique in the directory; however, the existing file should not be overwritten
ALLO	File name(s)	Allocate storage space for the files at the server
REST	File name(s)	Position the file marker at a specified data point
STAT	File name(s)	Return the status of files

FTP Response

Table 20.7 *Responses*

Code	*Description*
	Positive Preliminary Reply
120	Service will be ready shortly
125	Data connection open; data transfer will start shortly
150	File status is OK; data connection will be open shortly
	Positive Completion Reply
200	Command OK
211	System status or help reply
212	Directory status
213	File status
214	Help message
215	Naming the system type (operating system)
220	Service ready
221	Service closing
225	Data connection open
226	Closing data connection
227	Entering passive mode; server sends its IP address and port number
230	User login OK
250	Request file action OK

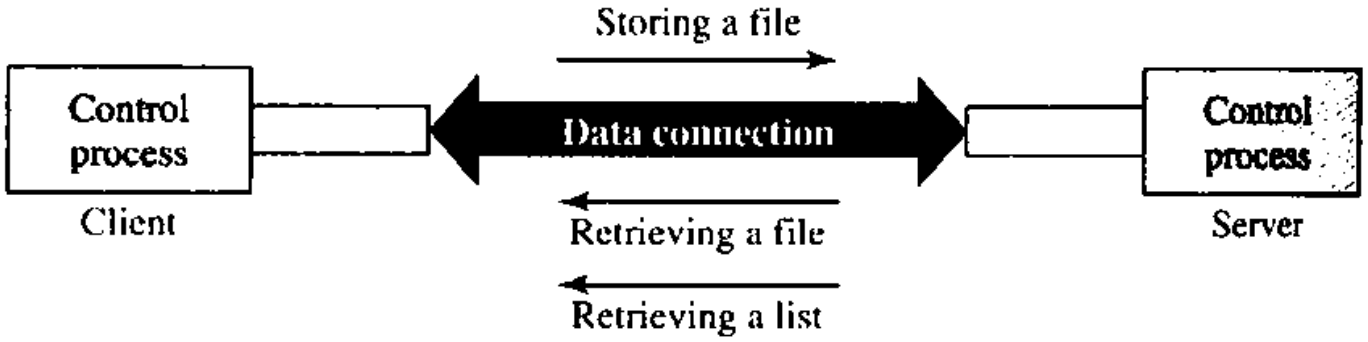
Table 20.7 *Responses (continued)*

Code	*Description*
	Positive Intermediate Reply
331	User name OK; password is needed
332	Need account for logging
350	The file action is pending; more information needed
	Transient Negative Completion Reply
425	Cannot open data connection
426	Connection closed; transfer aborted
450	File action not taken; file not available
451	Action aborted; local error
452	Action aborted; insufficient storage

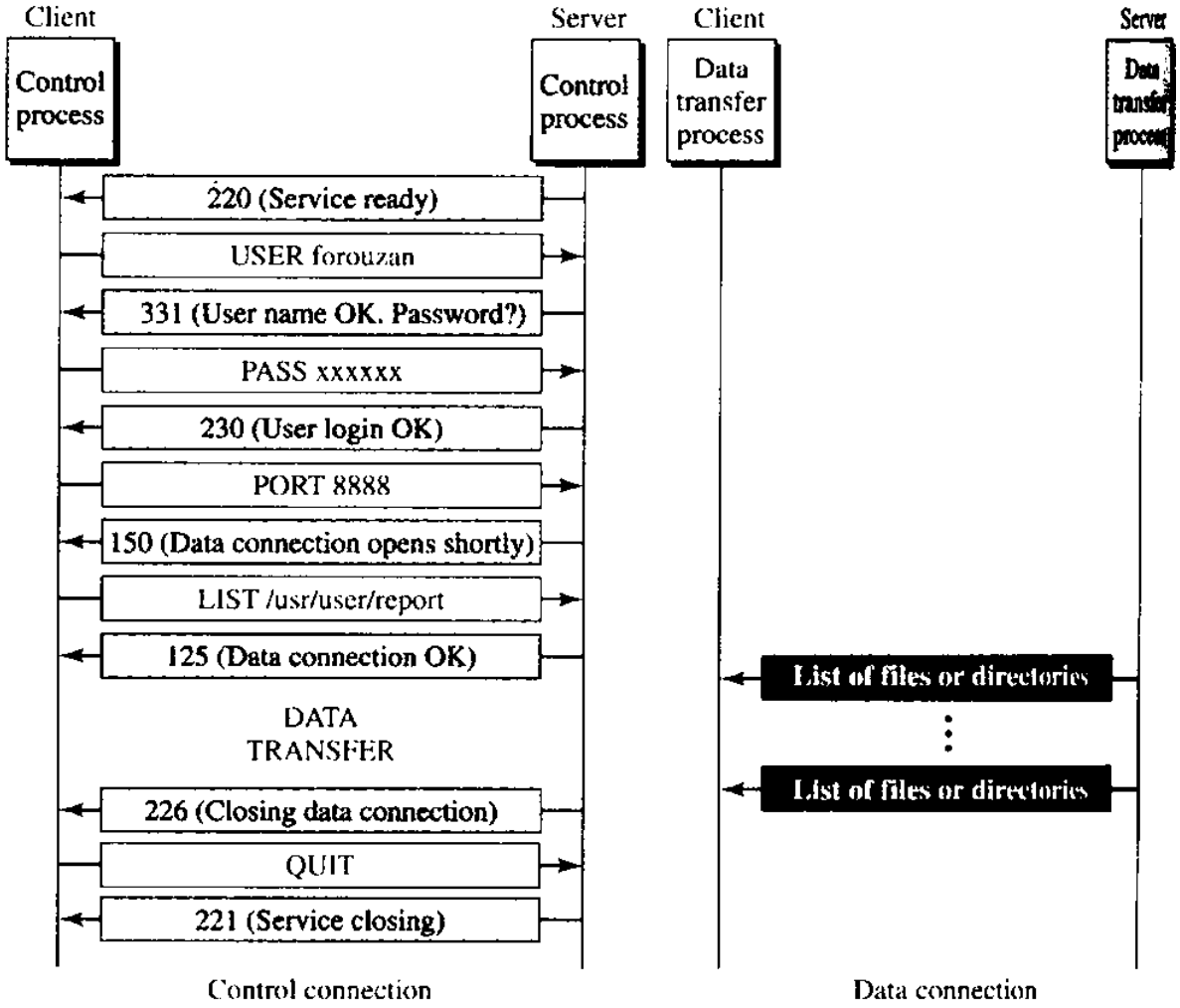
	Permanent Negative Completion Reply
500	Syntax error; unrecognized command
501	Syntax error in parameters or arguments
502	Command not implemented
503	Bad sequence of commands
504	Command parameter not implemented
530	User not logged in
532	Need account for storing file
550	Action is not done; file unavailable
552	Requested action aborted; exceeded storage allocation
553	Requested action not taken; file name not allowed

File Transfer

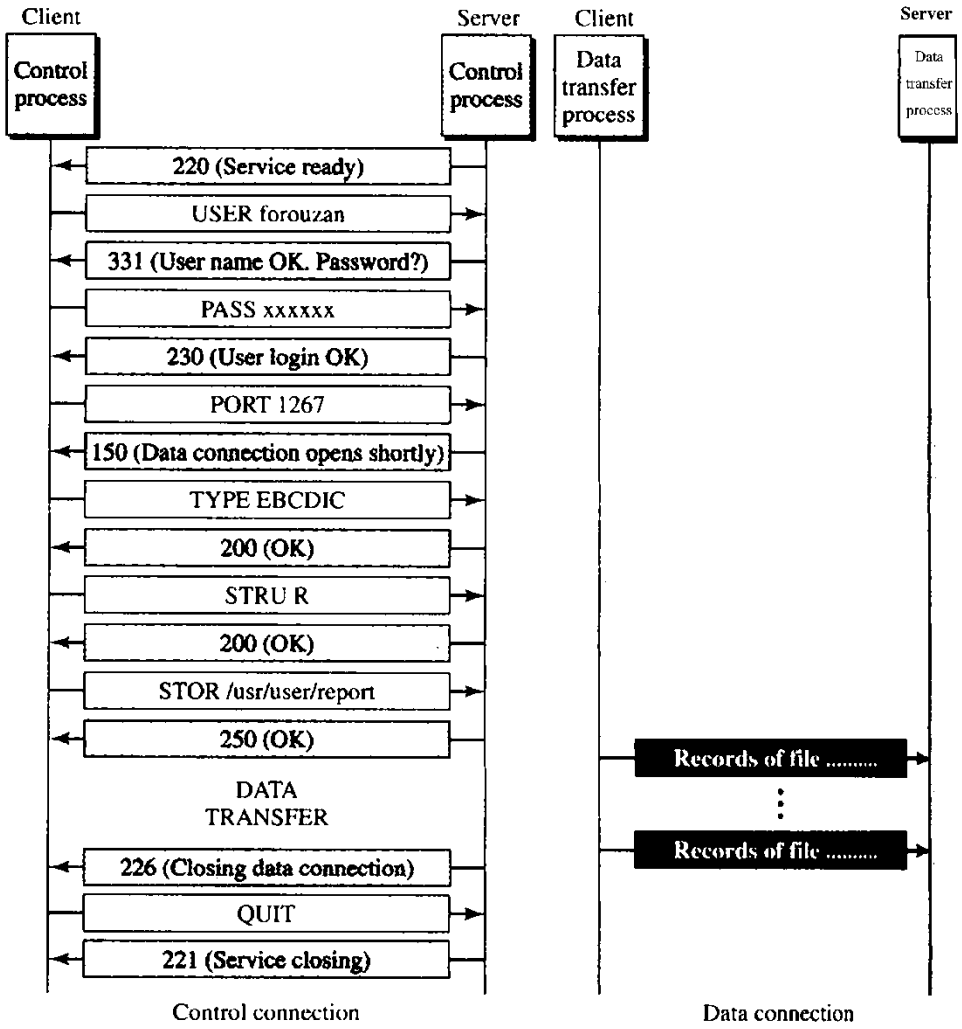
- **File transfer occurs over the data connection under the control of the commands set over the control connection.**



范例 1


Client
Control process
Server
Control process
Client
Data transfer process
Server
220 (Service ready)
USER forouzan
331 (User name OK. Password?)
PASS xxxxxx
230 (User login OK)
PORT 8888
150 (Data connection opens shortly)
LIST /usr/user/report
125 (Data connection OK)
DATA TRANSFER
226 (Closing data connection)
QUIT
221 (Service closing)
List of files or directories
List of files or directories
Control connection
Data connection

范例 2


Client
Server
Client
Server
Control process
Control process
Data transfer process
Data transfer process
220 (Service ready)
USER forouzan
331 (User name OK. Password?)
PASS xxxxxx
230 (User login OK)
PORT 1267
150 (Data connection opens shortly)
TYPE EBCDIC
200 (OK)
STRU R
200 (OK)
STOR /usr/user/report
250 (OK)
DATA TRANSFER
Records of file .....
Records of file .....
226 (Closing data connection)
QUIT
221 (Service closing)
Control connection
Data connection

Trivial File Transfer Protocol

- **TFTP(简单文件传输协议)：**
  - **TFTP uses the services of UDP on the well-known port 69.**
- **TFTP是一种简化的TCP/IP文件传输协议。TFTP只限于简单文件传输操作，它不提供权限控制，也不支持客户与服务器之间复杂的交互过程，没有庞大的命令集，没有列目录的功能，不能鉴别用户身份。因此TFTP软件比FTP软件小的多。**

TFTP

- TFTP 是一个很小且易于实现的文件传送协议。
- TFTP 使用客户服务器方式和使用 UDP 数据报，因此 TFTP 需要有自己的差错改正措施。
- TFTP 只支持文件传输而不支持交互。
- TFTP 没有一个庞大的命令集，没有列目录的功能，也不能对用户进行身份鉴别。

TFTP 的主要特点

- 每次传送的数据 PDU 中有 512 字节的数据，但最后一次可不足 512 字节。
- 数据 PDU 也称为文件块(block)，每个块按序编号，从 1 开始。
- 支持 ASCII 码或二进制传送。
- 可对文件进行读或写。
- 使用很简单的首部。

TFTP 的工作类似停止等待协议

- 发送完一个文件块后就等待对方的确认，确认时应指明所确认的块编号。
- 发完数据后在规定时间内收不到确认就要重发数据 PDU。
- 发送确认 PDU 的一方若在规定时间内收不到下一个文件块，也要重发确认 PDU。这样就可保证文件的传送不致因某一个数据报的丢失而告失败。

TFTP 的工作类似停止等待协议

- 在一开始工作时。TFTP 客户进程发送一个读请求 PDU 或写请求 PDU 给 TFTP 服务器进程，其熟知端口号码为 69。
- TFTP 服务器进程要选择一个新的端口和 TFTP 客户进程进行通信。
- 若文件长度恰为 512 字节的整数倍，则文件传送完毕后还必须最后发送一个只含首部而无数据的数据 PDU。
- 若文件长度不是 512 字节的整数倍，则最后传送数据 PDU 的数据字段一定不满512字节，这正好可作为文件结束的标志。

Network File System

- 网络文件系统（ Network File System ，NFS）
- NFS只是一种文件系统，本身没有传输功能，是基于RPC协议实现的，才能达到两个Linux系统之间的文件目录共享；

Network File System

- NFS 允许应用进程打开一个远地文件，并能在该文件的某一个特定的位置上开始读写数据。
- NFS 可使用户只复制一个大文件中的一个很小的片段，而不需要复制整个大文件。
- 例：计算机 A 的 NFS 客户软件，把要添加的数据和在文件后面写数据的请求一起发送到计算机 B 的 NFS 服务器。NFS 服务器更新文件后返回应答信息。
- 在网络上传送的只是少量的修改数据。

简单文件共享：SMB协议

- **Server Message Block**
- **基于TCP-NETBIOS下的，一般端口为139、445。**
  - **NetBIOS（网络基本输入/输出系统协议）协议是由IBM公司开发，主要用于数十台计算机的小型局域网。**
  - **NetBIOS协议是一种在局域网上的程序可以使用的API，为程序提供了请求低级服务的统一的命令集**
  - **几乎所有的局域网都是在NetBIOS协议的基础上工作的。**

字符界面的远程桌面连接

Telnet

远程终端协议 TELNET

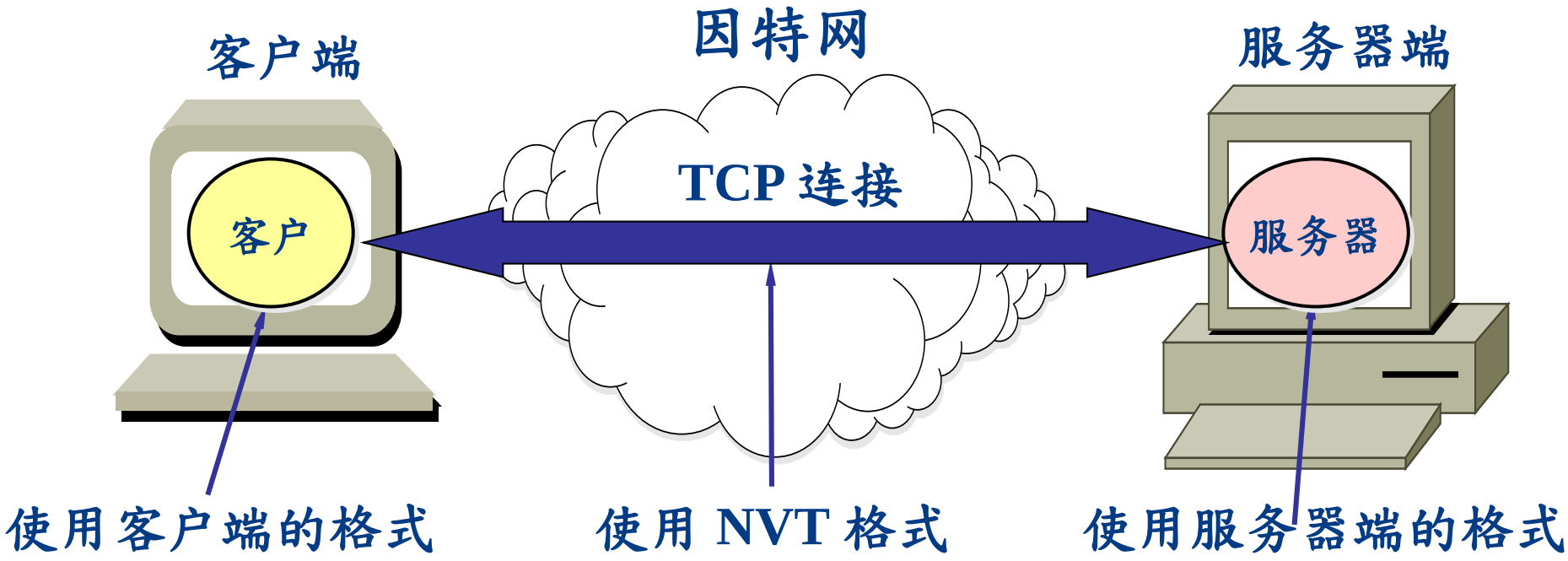
- TELNET 是一个简单的远程终端协议。
- 用户用 TELNET 就可在其所在地通过 TCP 连接注册（即登录）到远地的另一个主机上（使用主机名或 IP 地址）。
- TELNET 能将用户的击键传到远地主机，同时也能将远地主机的输出通过 TCP 连接返回到用户屏幕。这种服务是透明的，因为用户感觉到好像键盘和显示器是直接连在远地主机上。

客户–服务器方式

- 现在由于 PC 的功能越来越强，用户已较少使用 TELNET 了。（但在CUI界面系统仍使用）
- TELNET 也使用客户–服务器方式。在本地系统运行 TELNET 客户进程，而在远地主机则运行 TELNET 服务器进程。
- 和 FTP 的情况相似，服务器中的主进程等待新的请求，并产生从属进程来处理每一个连接。

TELNET 使用NVT 格式

- 网络虚拟终端 NVT



网络虚拟终端 NVT 格式

- 客户软件把用户的击键和命令转换成 NVT 格式，并送交服务器。
- 服务器软件把收到的数据和命令，从 NVT 格式转换成远地系统所需的格式。
- 向用户返回数据时，服务器把远地系统的格式转换为 NVT 格式，本地客户再从 NVT 格式转换到本地系统所需的格式。

PART IV Network Applications

Ch 32 World Wide Web Pages And Browsing

WWW页面与浏览

回顾：应用协议

应用层协议

- 协议类型
  - Private communication
    - 开发联机象棋程序
    - 不需要设计消息的协议内容
  - Standardized service
    - 开发FTP客户端
    - 需要遵循协议规范

应用层协议

- 关键方面
  - 数据表示
    - **Syntax of data items that are exchanged, specific form used during transfer, translation of integers, characters, and files between computers**
  - 数据传输
    - **Interaction between client and server, message syntax and semantics, valid and invalid exchange error handling, termination of interaction**

最后一个话题：网页

WWW, HTML, HTTP是一回事？

World Wide Web

- 三个关键标准
  - HyperText Language (HTML)
    - Markup A representation standard used to specify the contents and layout of a web page
  - Uniform Resource Locator (URL)
    - A representation standard that specifies the format and meaning of web page identifiers
  - HyperText Transfer Protocol (HTTP)
    - A transfer protocol that specifies how a browser interacts with a web server to transfer data

实例

- 运行于www.google.com，端口80
  - 客户端请求

```
GET / HTTP/1.1
Host:www.google.com
```

（紧跟着一个换行，通过敲入回车实现）

  - 服务器应答：

```
HTTP/1.1 200 OK
Content-Length: 3059
Server: GWS/2.0
Date: Sat, 11 Jan 2003 02:44:04 GMT
Content-Type: text/html
Cache-control: private
Set-Cookie:
PREF=ID=73d4aef52e57bae9:TM=1042253044:LM=1042253044:S=SMCc_HRPCQiqy
X9j; expires=Sun, 17-Jan-2038 19:14:07 GMT; path=/; domain=.google.com
Connection: keep-alive
```

    - （紧跟一个空行，并由HTML文本组成Google的主页）

HTML语言

- **HyperText Markup Language (HTML)**
- **Characteristics**
  - **Uses a textual representation**
  - **Describes pages that contain multimedia**
  - **Follows a declarative rather than procedural paradigm**
  - **Provides markup specifications instead of formatting**
  - **Permits a hyperlink to be embedded in an arbitrary object**
  - **Allows a document to include metadata**

HTML

- A language for creating Web pages.
- Beginning and ending tags
  - Beginning tag

```
<TagName Attribute=Value ... >
```

  - Ending tag

```
</TagName>
```

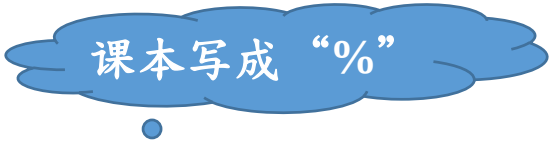
- Training: W3SCHOOL
  - http://www.w3school.com.cn

HTML program

```
<HTML>
        <HEAD>
                <TITLE>
                        text that forms the document title
                </TITLE>
        </HEAD>
        <BODY>
                body of the document appears here
        </BODY>
</HTML>
```

Uniform Resource Locator (URL)

- **The URL is a standard for specifying any kind of information on the Internet.**
- **The method is the protocol used to retrieve the document.**
- **URL:**




```
method://Host:Port/Path?Param
```

- **Example**

```
http://software.xmu.edu.cn:80/View/ArticleListView.aspx?group=1
```

Hypertext Transfer Protocol

- **The HyperText Transfer Protocol (HTTP) is the primary transfer protocol that a browser uses to interact with a web server.**
- **Characteristics**
  - **Uses textual control messages**
  - **Transfers binary data files**
  - **Can download or upload data**
  - **Incorporates caching**

Hypertext Transfer Protocol

- **HTTP is a protocol used mainly to access data on the World Wide Web.**
- **The protocol transfers data in the form of plain text, hypertext, audio, video, and so on.**
- **HTTP functions like a combination of FTP and SMTP.**
  - **HTTP uses only one TCP connection (well-known port 80).**
  - **The format of the messages is controlled by MIME-like headers.**

Request Methods

- **The request method is the actual command or request that a client issues to the server.**

Request	Description
GET	Requests a document; server responds by sending status information followed by a copy of the document
HEAD	Requests status information; server responds by sending status information, but does not send a copy of the document
POST	Sends data to a server; the server appends the data to a specified item (e.g., a message is appended to a list)

状态码 Status Codes

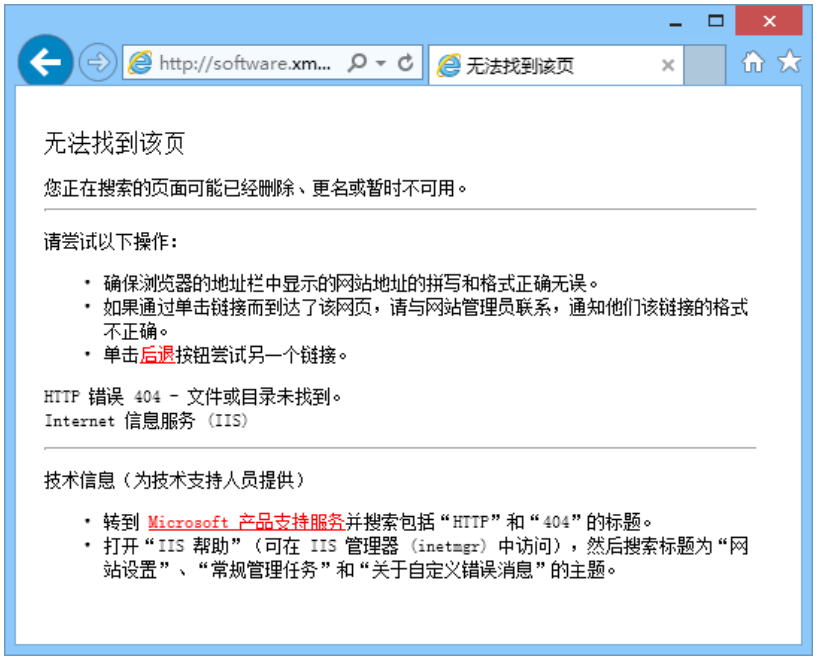
- **The status code field is similar to those in the FTP and the SMTP protocols.**

```
1xx Informational
100 Continue
101 Switching Protocols
...
2xx Success
200 OK
201 Created
202 Accepted
204 No Content
...
3xx Redirection
301 Moved Permanently
302 Found
304 Not Modified
...
4xx Client Error
400 Bad Request
401 Unauthorized
403 Forbidden
404 Not Found
405 Method Not Allowed
406 Not Acceptable
...
5xx Server Error
500 Internal Server Error
501 Not Implemented
503 Service Unavailable
...
```

Status Code

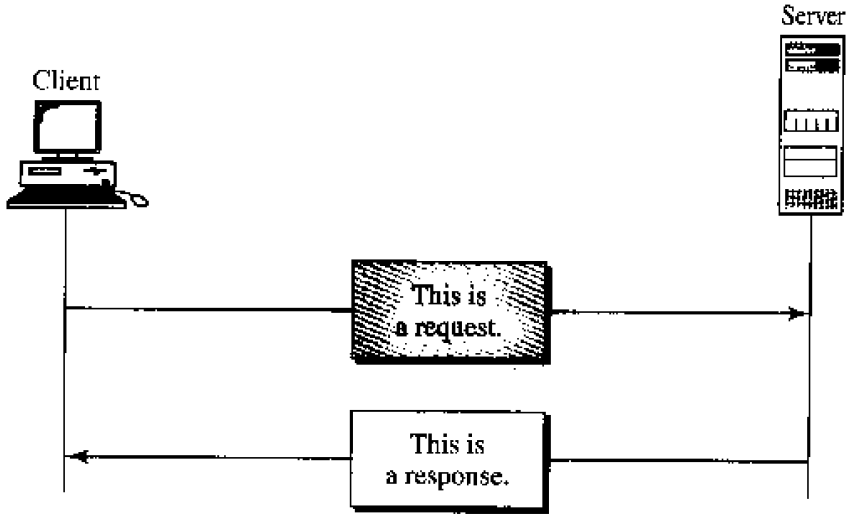
- 同样的Page Not Found，不一样的代码
  - 左图：服务器处理的重定向（HTTP 302）
  - 右图：服务器美化的错误页面（HTTP 404）





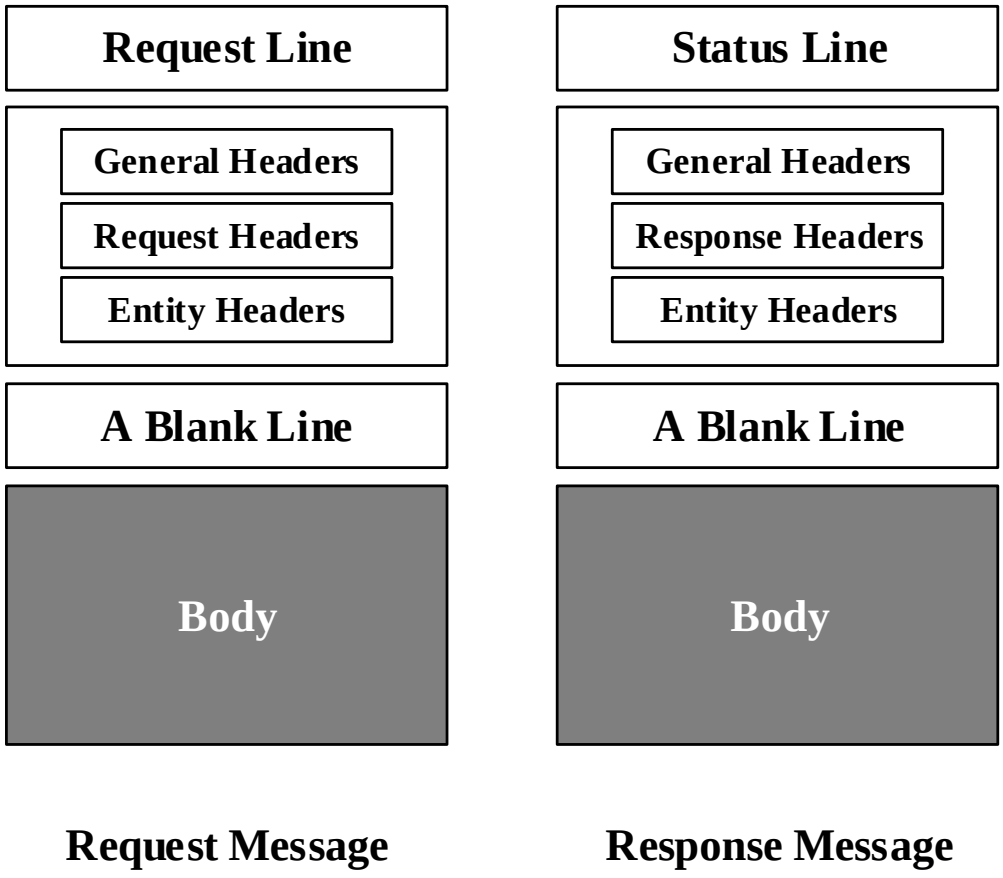
HTTP transaction

- **Although HTTP uses the services of TCP, HTTP itself is a stateless protocol**
- **HTTP使用六大对象**
  - **Response, Request, Server, Application, Session, Cookie**



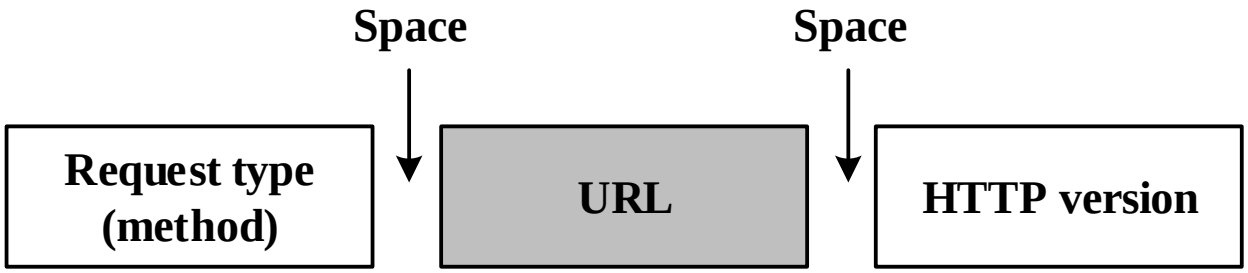
Message

- **There are two general types of HTTP messages:**



Request messages

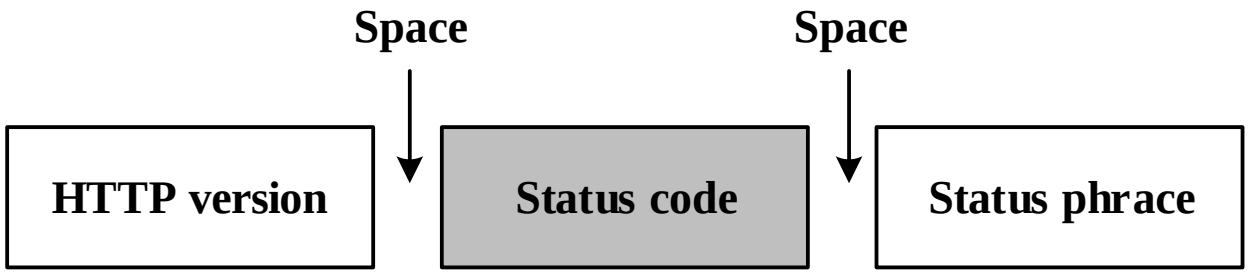
- **Request Line**



- **Headers**
- **A blank line**
- **Body**
  - **Present only in some messages**

Response message

- **Status Line**



- **Headers**
- **A blank line**
- **Body**
  - **Present only in some messages**

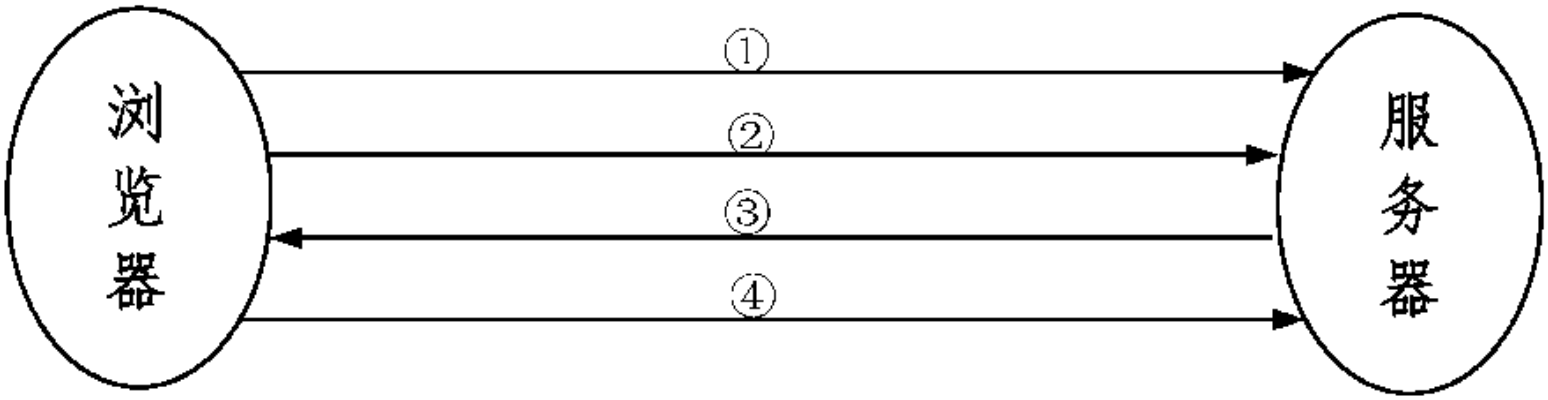
Headers

- **Header format** `HeaderName:<space>HeaderValue`
  - **exchanges additional info. between C-S.**
- **General headers**
  - **gives general information about the message and can be present in both a request and a response.**
- **Response headers**
  - **It specifies the server's configuration and special information about the request.**
- **Entity headers**
  - **It gives info. about the body of the document.**

缓存 Cache

- **A browser can reduce download times significantly by saving a copy of each image in a cache on the user's disk and using the cached copy.**
- **The `Last-Modified` header**
- **小文件不缓存**
  - **HEADER的时间和GET的时间差不多**

浏览器与服务器的通信过程



①浏览器与服务器连接 TCP 连接；

②浏览器向服务器发出 HTTP 请求，指定要传输的页面；

③服务器作出响应，包括状态码和页面文档；

④浏览器断开与服务器的 TCP 连接。

HTTP工作过程举例

- **URL：** `http://www.w3.org/protocols/HTTP-NG/Activity.html`
  - 浏览器分析链接指向页面的URL。
  - 浏览器向DNS请求解析www.w3.org的IP地址。
  - DNS解析出服务器的IP地址为18.23.0.23。
  - 浏览器与服务器建立TCP连接(18.23.0.23,端口80)
  - 浏览器发出取文件命令：GET /protocols/HTTP-NG/Activity.html。
  - www.w3.org服务器给出响应，将文件Activity.html发送给浏览器。
  - TCP连接释放。
  - 浏览器显示文件Activity.html中的所有文本。

Hypertext and Hypermedia

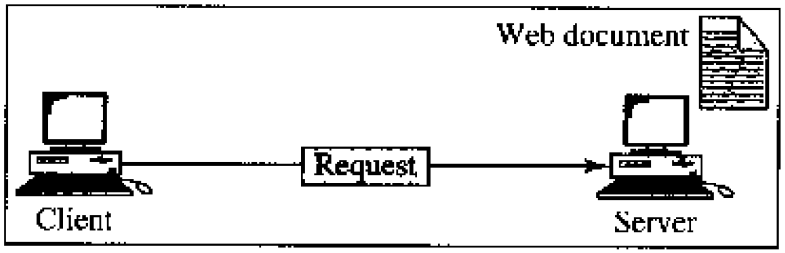
- **Hypertext documents contain only text.**
- **Hypermedia documents can contain pictures, graphics, and sound.**
- **The WWW uses the concept of hypertext and hypermedia.**
- **In a hypertext environment, information is stored in a set of documents that are linked together using the concept of pointers.**
- **An item can be associated with another document using a pointer.**

Categories of Web documents

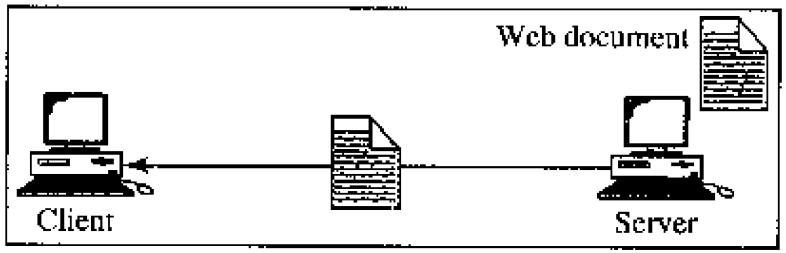
- 静态、动态和活动文档
- **The documents in the WWW can be grouped into three broad categories.**
  - **Static**
  - **Dynamic**
  - **Active**

Static document

- **Static documents**
  - **– fixed-content doc. created and stored in a sever**
  - **– The client can get only a copy of the document.**
  - **– Contents are determined when the file is created**
  - **– When a client accesses the document, a copy of the document is sent.**



a. Request

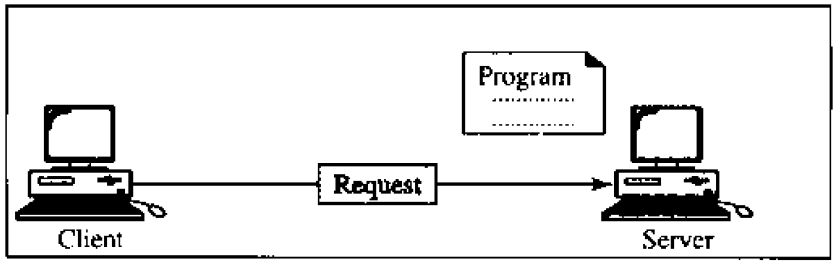


b. Response

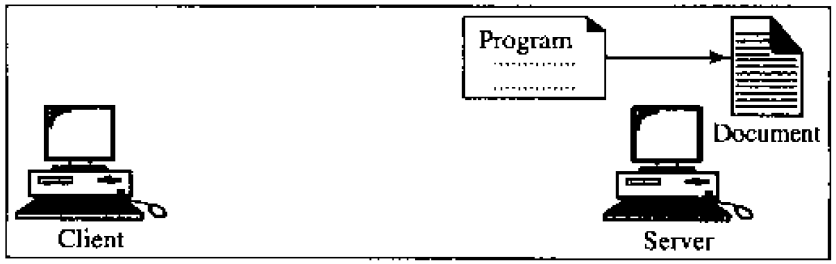
Dynamic document (1)

- A dynamic document is created by a Web server whenever a browser requests the document. When a request arrives, the Web server runs an application program that creates the dynamic document.
- The server returns the output of the program as a response to the browser that requested the document.
- Because a fresh document is created for each request, the contents of a dynamic document can vary from one request to another.

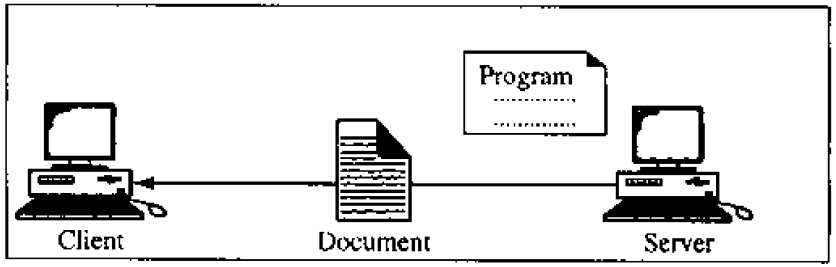
Dynamic document (2)



a. Request for running a program



b. Running the program and creating the document

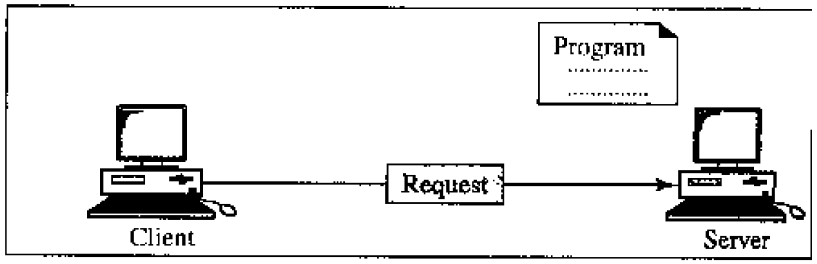


c. Response

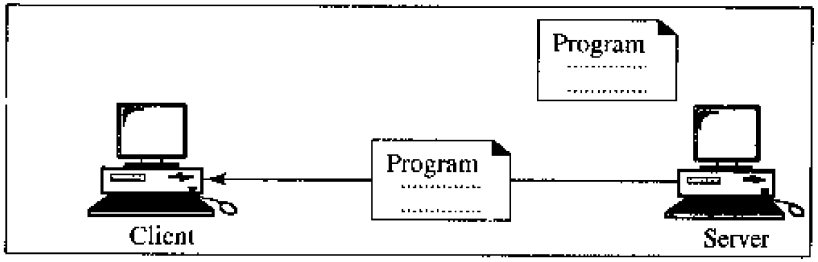
Active document (1)

- For many applications, we need a program to be run at the client site. There are called Active documents.
- When a browser requests an active document, the server sends a copy of the doc. in the form of byte code. The document is then run at the client (browser) site.
- An active document in the server is stored in the form of binary code.
- An active document is not run on the server.

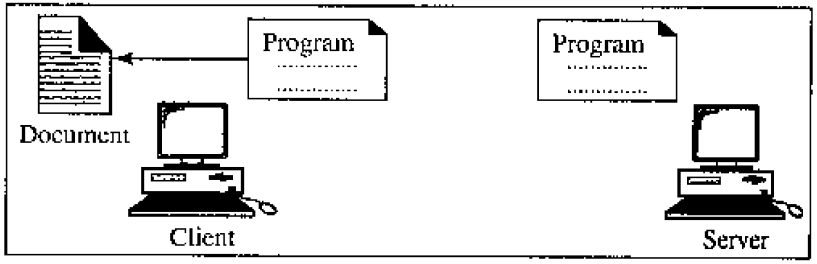
Active document (2)



a. Request for a copy of a program



b. Sending a copy of the program



c. Running the program and creating the document

计算机网络

18.

THANK YOU.



厦门大学软件学院

黄炜 助理教授